Longitudinal Data Analysis with R Statistical Software

Day 4. More on GEE and GLMM

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Overview of the Workshop

- Day 1: Introduction to Clustered and Longitudinal Data, simple analysis approaches
- Day 2: Generalized estimating equations (GEE)
- Day 3: Generalized linear mixed effects models (GLMM)
- **Day 4: More on GEE and GLMM**
- Day 5: Joint modeling of longitudinal and survival data

Day 3 Review

Comparison of GEE vs. GLM

- Marginal vs. conditional interpretation
- Model flexibility and assumptions
- Missing data
- Time dependent covariates

Marginal vs. Conditional Interpretation

- Parameter estimates obtained from a marginal model (as obtained via GEE) estimate population-averaged contrasts
- Parameter estimates obtained from a conditional model (as obtained via GLMM) estimate subject-specific contrasts
- In a linear model for a Gaussian outcome with an identity link, these contrasts are equivalent; not the case with non-linear models
    - Depends on the outcome distribution
    - Depends on the specified random effects

Marginal vs. Conditional Interpretation

Linear regression

- Conditional Model

  $E(Y|X_1, X_2) = \beta_0^\star + \beta_1^\star X_1 + \beta_2^\star X_2$

  - β_1 has conditional interpretation
  - Conditioning on X_2, for a 1-unit increase in X_1 the expectation of Y (holding X_2 constant) increases $\beta_1^\star$.

- Marginal Model

  $E(Y|X_1) = \beta_0 + \beta_1 X_1$

  - β_1 has a marginal (unconditional) interpretation
  - For a 1-unit increase in X_1 the expectation of Y increases β_1.

Marginal vs. Conditional Interpretation

Linear regression (continued)

- Conditional Model

  $E(Y|X_1, X_2) = \beta_0^\star + \beta_1^\star X_1 + \beta_2^\star X_2$

- Marginal Model

  $E(Y|X_1) = \beta_0 + \beta_1 X_1$

- Computing Marginal from Conditional Model

$$
\begin{aligned}
E(Y|X_1) &= E[E(Y|X_1, X_2)] \\
&= E[\beta_0^\star + \beta_1^\star X_1 + \beta_2^\star X_2] \\
&= \beta_0^\star + \beta_1^\star X_1 + \beta_2^\star E(X_2|X_1).
\end{aligned}
$$

Marginal vs. Conditional Interpretation

Longitudinal Data

- Linear Mixed Effects Model

  $E(Y_{ij}|X_{ij}) = \beta_0^\star + \beta_1^\star X_{ij} + b_{i0}$

  - Conditional on random effect b_{i0}
  - Correlation handled with random intercept

- Generalized Estimating Equations Linear Model

  $E(Y_{ij}|X_{ij}) = \beta_0 + \beta_1 X_{ij}$

  - Marginal model
  - Correlation handled in estimating equations but not directly in model

Marginal vs. Conditional Interpretation

Longitudinal Data with another Covariate

- Linear Mixed Effects Model

  $$E(Y_{ij}|X_{ij}, V_i) = \beta_0^\star + \beta_1^\star X_{ij} + \beta_2^\star V_i + b_{i0}$$

  - Still referred to as a 'Conditional Model' because conditioning on random effect

- Generalized Estimating Equations Linear Model

  $$E(Y_{ij}, V_i|X_{ij}) = \beta_0 + \beta_1 X_{ij} + \beta_2 V_i$$

  - Still referred to as a 'Marginal Model' because not conditioning on random effect

Marginal vs. Conditional Interpretation

- Linear Mixed Effects Model (Conditional)

  $$E(Y_{ij}|X_{ij}) = \beta_0^\star + \beta_1^\star X_{ij} + b_{i0}$$

- Generalized Estimating Equations Linear Model (Marginal)

  $$E(Y_{ij}|X_{ij}) = \beta_0 + \beta_1 X_{ij}$$

- Computing Marginal from Conditional Model

$$\begin{aligned} E(Y_{ij}|X_{ij}) &= E[E(Y_{ij}|X_{ij}, b_{i0})] \\ &= E[\beta_0^\star + \beta_1^\star X_{ij} + b_{i0}] \\ &= \beta_0^\star + \beta_1^\star X_{ij} + E(b_{i0}|X_{ij}) \\ &= \beta_0^\star + \beta_1^\star X_{ij} \\ &= \beta_0 + \beta_1 X_{ij}. \end{aligned}$$

  - With linear models, GLMM and GEE yield the same results.

Marginal vs. Conditional

GLMM and GEE do not generally give the same results.

Example

Consider a logistic regression model with subject-specific intercepts

$$\text{logit}(P[Y_{ij} = 1 \mid b_{i0}]) = \beta_0^\star + \beta_1^\star x_{ij} + b_{i0}.$$

- The expectation is

$$\begin{aligned} E[Y_{ij}|x_{ij}, b_{i0}] &= P[Y_{ij} = 1|x_{ij}, b_{i0}] \\ &= \frac{\exp(\beta_0^\star + \beta_1^\star x_{ij} + b_{i0})}{1 + \exp(\beta_0^\star + \beta_1^\star x_{ij} + b_{i0})} \end{aligned}$$

- Assuming $\{\beta_0^\star, \beta_1^\star\} = \{-2, 0.4\}$, then the conditional exposure odds ratio is

$$\exp(\beta_1^\star) = \exp(0.4) = 1.49$$

Example (continued)

The marginal (population) expectation conditional on x_{ij} across individuals is

$$
\begin{aligned}
P[Y_{ij} = 1 | x_{ij}] &= \int P[Y_{ij} = 1 \mid x_{ij}, b_{i0}] dF(b_{i0}) \\
&= \int \frac{\exp(\beta_0^\star + \beta_1^\star x_{ij} + b_{i0})}{1 + \exp(\beta_0^\star + \beta_1^\star x_{ij} + b_{i0})} dF(b_{i0}),
\end{aligned}
$$

where typically $b_{i0} \sim N(0, D)$.

- Assuming $\{\beta_0^\star, \beta_1^\star\} = \{-2, 0.4\}$ and $D = 2$ the population expectations (via integration) are

$$
\begin{aligned}
P[Y_{ij} = 1 \mid x_{ij} = 0] &= 0.18 \\
P[Y_{ij} = 1 \mid x_{ij} = 1] &= 0.23.
\end{aligned}
$$

Example (continued)

A marginal model ignores heterogeneity among individuals and considers the population-averaged rate rather than the conditional rate, fitting this model:

$$\text{logit}(P[Y_{ij} = 1 | x_{ij}]) = \beta_0 + \beta_1 x_{ij}.$$

Based on the true (conditional) model,

$$\begin{aligned} P[Y_{ij} &= 1 \mid x_{ij} = 0] = 0.18 \\ P[Y_{ij} &= 1 \mid x_{ij} = 1] = 0.23 \end{aligned}$$

and the population-averaged odds ratio associated with exposure is

$$\frac{P[Y_{ij} = 1 \mid x_{ij} = 1]/(1 - P[Y_{ij} = 1 \mid x_{ij} = 1])}{P[Y_{ij} = 1 \mid x_{ij} = 0]/(1 - P[Y_{ij} = 1 \mid x_{ij} = 0])} = \frac{0.23/(1-0.23)}{0.18/(1-0.18)} = 1.36$$

so that $\{\beta_0, \beta_1\} = \{\text{logit}(0.18), \log(1.36)\} = \{-1.52, 0.31\}$.

- Marginal parameters $\{-1.52, 0.31\}$ are 'attenuated' compared with conditional parameters $\{-2.0, 0.4\}$.

Marginal vs. Conditional

Interpretation of GLMM:

		Fitted model	
Outcome	Coefficient	Random intercept	Random intercept/slope
Continuous	Intercept	Marginal	Marginal
	Slope	Marginal	Marginal
Count	Intercept	Conditional	Conditional
	Slope	Marginal	Conditional
Binary	Intercept	Conditional	Conditional
	Slope	Conditional	Conditional

⋆ Marginal = population-averaged; conditional = subject-specific

Marginal vs. Conditional

Interpretation

- Some people (including me, BS) do not like the coefficient interpretation to depend on the random effect, which is an unknown latent variable that is different per person.
  - The effect is within a person.
  - Results and interpretation depend on choice of random effects distribution.
- Some people (including me, BS) also like the model interpretation with a random effects.
  - Sometimes we want to learn about the random effects.
  - Random effects describe variance and heterogeneity in the data.
- But it is hard to have it both ways
  - Marginalized models (e.g., Heagerty PJ, 1999; Biometrics 55: 688–98)

Model Flexibility and Assumptions

Linear mixed effects models:

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + b_{i0} + \epsilon_{ij}.$$

- Typically assume a full distribution and fit maximum likelihood estimates.
- Requires assuming distribution of ϵ_{ij} and distribution of b_{i0}.
- Often normality is assumed, occasionally correlation structure between ϵ_{ij} and $\epsilon_{ij'}$ assumed.
- Models are not always robust to misspecification.
- Methods continue to be developed to make these models less parametric and more flexible.

Model flexibility and assumptions

Generalized estimating equations with a linear model:

$$E(Y_{ij}|X_{ij}) = \beta_0 + \beta_1 X_{ij}.$$

- Does not require assuming a full distribution.
- Only makes assumption on the mean relationship (first moment follows linear model).
- Working correlation can be misspecified, but estimates are consistent and coverage probability correct.
- So generally more robust than linear mixed effects models.
- But, makes stronger assumptions than LMMs with missing data.

Missing Data

- Missing values arise in longitudinal studies whenever the intended serial observations collected on a subject over time are incomplete
  - Collect fewer data than planned $\Rightarrow$ decreased efficiency (power)
  - Missingness can depend on outcome values $\Rightarrow$ potential bias (generalizability)
- Important to distinguish between missing data and unbalanced data, although missing data necessarily result in unbalanced data
- Missing data require consideration of the factors that influence the missingness of intended observations
- Also important to distinguish between intermittent missing values (non-monotone) and dropouts in which all observations are missing after dropout (monotone)

Pattern	t_1	t_2	t_3	t_4	t_5
Monotone	3.8	3.1	2.0	$\boxed{}$	$\boxed{}$
Non-monotone	4.1	$\boxed{}$	3.8	$\boxed{}$	$\boxed{}$

Missing Data Mechanisms

Partition the complete set of intended observations into the observed and missing data; what factors influence missingness of intended observations?

- **Missing completely at random** (MCAR)
  - Missingness does not depend on **either** the observed or missing data
- **Missing at random** (MAR)
  - Missingness depends **only** on the observed data
- **Missing not at random** (MNAR)
  - Missingness depends on **both** the observed and missing data

MNAR also referred to as informative or non-ignorable missingness; thus MAR and MCAR as non-informative or ignorable missingness (Rubin, 1976)

Examples and implications

- **MCAR**: Administrative censoring at a fixed calendar time
  - Generalized estimating equations are valid
  - Mixed-effects models are valid
- **MAR**: Individuals with no current weight loss in a weight-loss study
  - Generalized estimating equations are not valid
  - Mixed-effects models are valid
- **MNAR**: Subjects in a prospective study based on disease prognosis
  - Generalized estimating equations are not valid
  - Mixed-effects models are not valid

⋆ MAR and MCAR can be evaluated (in part) using the observed data

Potential Solutions for Outcome Missing Data

- Complete case analysis
  - OK if MCAR
  - OK if MAR and fitting mixed-effects model
  - Biased if MAR and fitting GEE
- Last observation carried forward
  - Bad idea always
- Single imputation
  - Under-estimates uncertainty
- Inverse probability weighting (IPW)
  - Can be used to address MAR with GEE
  - Generally no need to use to address MAR with mixed-effects models
- Multiple imputation (MI)
  - Can be used to address MAR with GEE
  - Generally no need to use to address MAR with mixed-effects models

⋆ **With MAR covariates**, IPW and MI can be used and are generally beneficial for both GEE and mixed-effects models

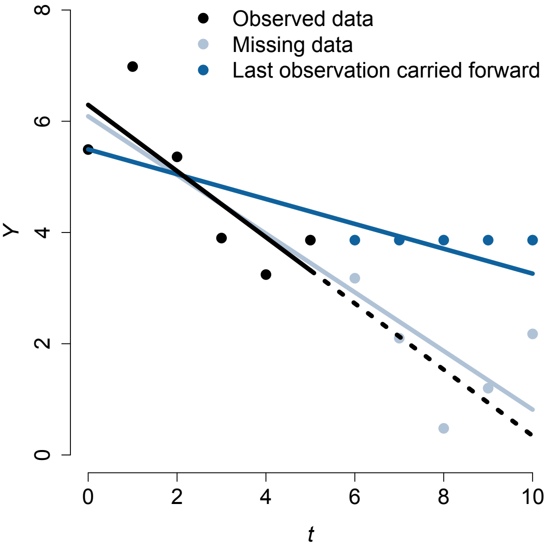
Last observation carried forward

- Extrapolate the last observed measurement to the remainder of the intended serial observations for subjects with any missing data

ID	t_1	t_2	t_3	t_4	t_5
1	3.8	3.1	2.0	$\boxed{2.0}$	$\boxed{2.0}$
2	4.1	3.5	3.8	2.4	2.8
3	2.7	2.4	2.9	3.5	$\boxed{3.5}$

- May result in serious bias in either direction
- May result in anti-conservative p-values; variance is understated
- Has been thoroughly repudiated, but still a standard method used by the pharmaceutical industry and appears in published articles
- A refinement would extrapolate based on a regression model for the average trend, which may reduce bias, but still understates variance

Last observation carried forward



Multiple Imputation

- Multiple imputation is a popular approach for handling missing at random data
- Key component of the 2025 Vanderbilt-Nigeria Biostatistics Training Workshop
- Basic idea:
    1. Fit imputation model to the observed data that will be used to predict (impute) values of the missing data
    2. Impute missing data from this model, incorporating uncertainty (e.g., don't just impute predicted means)
    3. With the complete (imputed + observed) data, fit analysis model and store estimates of interest
    4. Repeat steps 2-3 multiple times (e.g., 5-50)
    5. Pool the stored estimates by taking the average and compute their variance (usually using Rubin's rules)

Multiple Imputation

- Fit imputation model to observed data and impute missing data from that model

ID	t_1	t_2	t_3	t_4	t_5
1	3.8	3.1	2.0	$\boxed{2.3}$	$\boxed{1.7}$
2	4.1	3.5	3.8	2.4	2.8
3	2.7	2.4	2.9	3.5	$\boxed{3.0}$

- Fit analysis model to complete data
- Repeat: Impute missing data from imputation model

ID	t_1	t_2	t_3	t_4	t_5
1	3.8	3.1	2.0	$\boxed{1.8}$	$\boxed{2.5}$
2	4.1	3.5	3.8	2.4	2.8
3	2.7	2.4	2.9	3.5	$\boxed{2.7}$

- Fit analysis model to complete data
- Combine estimates across multiple repetitions

Multiple Imputation

- Building an imputation model
  - `mice` function in R is good for multiple imputation
  - Wide versus long format, could impact the imputation model
- Wide Format

ID	sex	t_1	t_2	t_3	t_4	t_5
1	M	3.8	3.1	2.0	NA	NA
2	M	4.1	3.5	3.8	2.4	2.8
3	F	2.7	2.4	2.9	3.5	NA

- Imputation model uses information from other times within same person to multiply impute missing values

Multiple Imputation

- Long Format

ID	sex	time	outcome
1	M	1	3.8
1	M	2	3.1
1	M	3	2.0
2	M	1	4.1
2	M	2	3.5
2	M	3	3.8
2	M	4	2.4
2	M	5	2.8
3	F	1	2.7
3	F	2	2.4
3	F	3	2.9
3	F	4	3.5

- The computer does not realize there is any missing data to impute.

Multiple Imputation

- Long Format with missing values noted.

ID	sex	time	outcome
1	M	1	3.8
1	M	2	3.1
1	M	3	2.0
1	M	4	NA
1	M	5	NA
2	M	1	4.1
2	M	2	3.5
2	M	3	3.8
2	M	4	2.4
2	M	5	2.8
3	F	1	2.7
3	F	2	2.4
3	F	3	2.9
3	F	4	3.5
3	F	5	NA

- Default would fit an imputation model that predicts missing values using ID, sex, and time, but does not explicitly incorporate the other outcome values from the same person.

Time Dependent Exposures

Important analytical issues arise with time-dependent exposures

1. May be necessary to correctly specify the lag relationship over time between outcome $y_i(t)$ and exposure $x_i(t)$, $x_i(t-1)$, $x_i(t-2), \ldots$ to characterize the underlying biological latency in the relationship
   - **Example**: Air pollution studies may examine the association between heart attack on day t and pollutant levels on days t, $t-1$, $t-2, \ldots$
2. May exist exposure endogeneity in which the outcome at time t predicts the exposure at times $t' > t$; motivates consideration of alternative targets of inference and corresponding estimation methods
   - **Example**: If $y_i(t)$ is a symptom measure and $x_i(t)$ is an indicator of drug treatment, then past symptoms may influence current treatment

Time Dependent Exposure Definitions

Factors that influence $x_i(t)$ require consideration when selecting analysis methods to relate a time-dependent exposure to longitudinal outcomes

- **Exogenous**: An exposure is exogenous w.r.t. the outcome process if the exposure at time t is conditionally independent of the history of the outcome process $\mathcal{Y}_i(t) = \{y_i(s) \mid s \leq t\}$ given the history of the exposure process $\mathcal{X}_i(t) = \{x_i(s) \mid s \leq t\}$

$$[x_i(t) \mid \mathcal{Y}_i(t), \mathcal{X}_i(t)] = [x_i(t) \mid \mathcal{X}_i(t)]$$

- **Endogenous**: Not exogenous

$$[x_i(t) \mid \mathcal{Y}_i(t), \mathcal{X}_i(t)] \neq [x_i(t) \mid \mathcal{X}_i(t)]$$

Time Dependent Exposure Examples

Exogeneity may be assumed based on the design or evaluated empirically

- **Observation time**: Any analysis that uses scheduled observation time as a time-dependent exposure can safely assume exogeneity because time is "external" to the system under study and thus not stochastic
- **Cross-over trials**: Although treatment assignment over time is random, in a randomized study treatment assignment and treatment order are independent of outcomes by design and therefore exogenous
- **Other examples**: Weather patterns, geopolitical events (e.g., COVID-19 pandemic)
- **Empirical evaluation**: Endogeneity may be empirically evaluated using the observed data by regressing current exposure $x_i(t)$ on previous outcomes $y_i(t-1)$, adjusting for previous exposure $x_i(t-1)$

$$g(\mathrm{E}[X_i(t)]) = \theta_0 + \theta_1 y_i(t-1) + \theta_2 x_i(t-1)$$

and using a model-based test to evaluate the null hypothesis: $\theta_1 = 0$

Time Dependent Exposure Implications

The presence of endogeneity determines specific analysis strategies

- If exposure is exogenous, then the analysis can focus on specifying the lag dependence of $y_i(t)$ on $x_i(t)$, $x_i(t-1)$, $x_i(t-2), \ldots$
- If exposure is endogenous, then analysts must focus on selecting a meaningful target of inference and valid estimation methods, which are often more complicated

Time Dependent Exposures, Targets of Inference

With longitudinal outcomes and a time-dependent exposure there are several possible conditional expectations that may be of scientific interest

- **Fully conditional model**: Include the entire exposure process

$$E[Y_i(t) \mid x_i(1), x_i(2), \ldots, x_i(T_i)]$$

- **Partly conditional models**: Include a subset of exposure process

$$E[Y_i(t) \mid x_i(t)]$$
$$E[Y_i(t) \mid x_i(t-k)] \text{ for } k \leq t$$
$$E[Y_i(t) \mid \mathcal{X}_i(t) = \{x_i(1), x_i(2), \ldots, x_i(t)\}]$$

⋆ An appropriate target of inference that reflects the scientific question of interest must be identified prior to selection of an estimation method

Time Dependent Exposures, Key Assumption

Suppose that primary scientific interest lies in a cross-sectional mean model

$$E[Y_i(t) \mid x_i(t)] = \beta_0 + \beta_1 x_i(t)$$

To ensure consistency of a generalized estimating equation or likelihood- based mixed-model estimator for β, it is sufficient to assume that

$$E[Y_i(t) \mid x_i(t)] = E[Y_i(t) \mid x_i(1), x_i(2), \ldots, x_i(T_i)]$$

Otherwise an independence estimating equation should be used

- Known as the full covariate conditional mean assumption
- Implies that with time-dependent exposures must assume exogeneity when using a covariance-weighting estimation method
- The full covariate conditional mean assumption is often overlooked and should be verified as a crucial element of model verification

Summary of Day 4

Comparison of GEE vs. GLM

- Marginal vs. conditional interpretation
- Model flexibility and assumptions
- Missing data

Time dependent covariates